AI: Strategy + Marketing (MGT 853)

ML Essentials (Session 2)

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Definitions: AI and ML

Artificial Intelligence

"...Intelligence can in principle be so precisely described that a machine can be made to simulate it." (John McCarthy)

Machine Learning

- "The field of study that gives computers the ability to learn without explicitly being programmed" (Arthur Samuel)

What's the difference?

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Machine Learning

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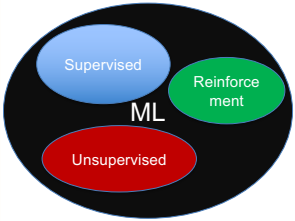
- "Improve over Task T with resect to some performance measure P based on experience E" (Tom Mitchell)

What's the difference?

Differences between ML and other AI approaches

- ML primarily is "learning from data"



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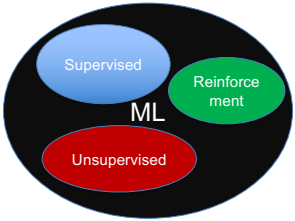

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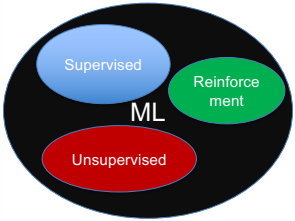

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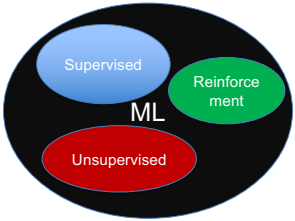

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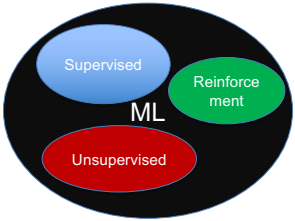

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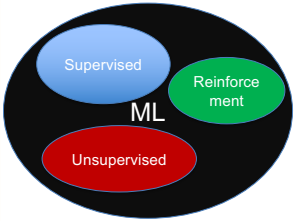

Rest of the course will focus on ML (use interchangeably with AI)

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- Integrating Prediction with decision making can be challenging (Module B)

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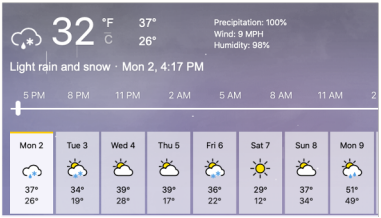
Most commonly used form of ML in practice



Logistic Regression
Polynomial Regression
Support Vector Machines
Decision Trees
Deep Neural Nets

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Can use for exploratory analysis and segmentation even when question is unclear

Cluster Analysis
K-means
K-Nearest Neighbor
Association Rule Mining
Principal Components Analysis

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- Building an ML model in Colab
- Before the model (Pre-processing)

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- After the Model (Post-processing)

ML Pipeline

Data

Training, Validation and Test

Why do we need to split data?

Peeking is a consequence of using test-set performance to both choose a hypothesis and evaluate it. The way to avoid this is to really hold the test set out – *lock it away* until you are completely done with learning and simply wish to obtain an independent evaluation of the final hypothesis. (And then, if you don't like the results . . . you have to obtain, and lock away, a completely new test set if you want to go back and find a better hypothesis.)

– **Russell and Norvig (Artificial Intelligence: A Modern Approach)**

Data

Training, Validation and Test

Why do we need to split data?

If the test set is locked away, but you still want to measure performance on unseen data as a way of selecting a *good hypothesis*, then divide the available data (without the test set) into a training set and a validation set.

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Pre-processing

Why?

What is Data Pre-processing?

Transforming the Raw Data to a format that is suitable for use in an ML algorithm. Two types of pre-processing: Lossy and Lossless

- Identify anomalies in the data (missing values, duplicates, outliers, inconsistent or implausible values etc.)

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- Consistent format of possibly acquired from multiple sources
- Reduce algorithmic bias (How?)

Feature Engineering

What is Feature Engineering?

The process of identifying specific features, transforming them, or *creating new features* from data is called feature engineering.

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- Lots of possibilities with unstructured data

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- For Random Forest hyperparameters, see: `https://scikit-learn.org/stable/modules/generated/sklearn.`

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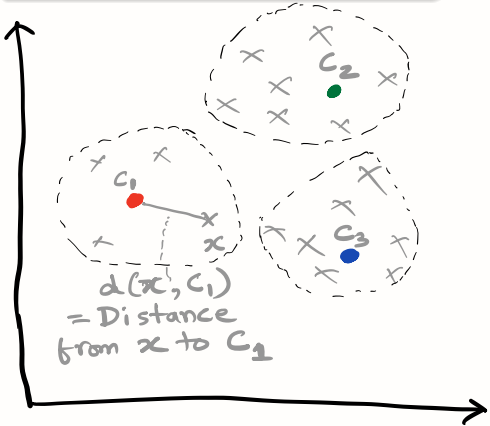
- But the question is do they "generalize" such performance when

K-means

What is K-means?

Group similar data together



- Group data so that points are close to the center of their clusters, but far away from other clusters
- Let's say the center of cluster i is C_i
- Any point x in the cluster has a distance of $d(x, C_i) = (x - C_i)$ from the center
- How do you decide which cluster a point "belongs to"? See which center is the closest to the point.
- There are many such points x in the cluster so we sum over all of them

K-means

- Details at: `https://towardsdatascience.com/the-math-and-code-behind-k-means-clustering-795582423666`
- Probably the most common unstructured ML method used
- Typically used for customer segmentation (finding "similar" groups of customers)
- K is a **hyperparameter** chosen based on the business application
  - Typically $K = 2, 3, 4$ in applications
- Can devise different interventions for each group
- Other Applications: `https://dzone.com/articles/10-interesting-use-cases-for-the-k-means-algorithm`

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- ML methods have many challenges including underfitting / overfitting
- Need to carefully choose appropriate methods for your problem